Ruşen Birben

AI Researcher & Engineer

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AI and Data Engineering graduate from Istanbul Technical University and independent researcher with METU's Applied NLP Group. Research interests center on LLM reliability and evaluation, with experience in numerical precision in interpretability, Turkish LLM pre-training data, Turkish NLP benchmarks and datasets, machine unlearning, and LLM applications for finance.

EDUCATION

B.Sc. AI and Data Engineering
2022 - 2025

Istanbul Technical University
GPA: 3.61 / 4.00

Graduated July 2025

Electronics and Communication Engineering
2020 - 2022

Istanbul Technical University

Transferred to AI & Data Engineering

PUBLICATIONS & RESEARCH CONTRIBUTIONS

Patching Fails Small, Attribution Fails Large: Component Effects in 16-Bit Language Models
2026

Transactions on Machine Learning Research (TMLR) - Under review

- Sole author.
- Shows across ten language models that bfloat16 puts an error floor under intervention-based interpretability effects, while attribution keeps the ranking and sign of single-neuron effects (rank correlation ≥ 0.97); recommends measuring the floor before trusting 16-bit interventions.

Tokenization Strategies for Pretraining Generative Large Language Models: A Case Study in Turkish
2026

Journal of Artificial Intelligence Research (JAIR) - Under review

- Evren Ayberk Munis and **Ruşen Birben** (equal contribution, joint first authors), Çağrı Toraman.
- Compared eight lossless tokenizers, including morpheme-constrained BPE, for Turkish LLM pretraining on a fixed 30 GB budget: bits per byte differed by at most 1.2%, while estimated training compute differed by up to 49%; morpheme-constrained BPE cost 23 to 29% more compute than byte-level BPE.

Exact First, Then Fast: An Event-for-Event C++ Rebuild of the Agenthon ABIDES Market
2026

Agenthon Workshop @ NeurIPS 2026 - In-person poster, Atlanta, USA, December 12, 2026 Competition

- Göktürk Batın Dervişoğlu, Muhammed Faruk Aytin, **Muhammed Rüşen Birben**.
- Rebuilt the Agenthon ABIDES market simulator in C++, preserving event ordering, random-number sequences, and reference outputs. Validated exact agreement across 6,900 randomized comparisons and achieved a 1,299× simulation-loop speedup on the largest public benchmark.

Turkish Cultural Knowledge Dataset
2026

MRL Workshop @ EMNLP 2026 - Dataset accepted; joint paper in preparation Shared Task

- Led the source-grounded agent pipeline and data preparation for a Turkish cultural knowledge dataset spanning 15 domains.
- Generated 1,200 candidate question-answer pairs with 15 domain agents, selected 500 for sociological review by Büşra Tufan (who chose the final 300), and prepared the submission.
- Reviewed English translations; supervised by Çağrı Toraman.

SCHOLARSHIPS & AWARDS

1416 YLSY Scholarship · M.Sc. & Ph.D. 2026

Fully funded government scholarship for M.Sc. and Ph.D. study in Artificial Intelligence and Big Data in the USA, awarded through a national selection exam.

1st Place · Agenthon 2026, Track 3 (Simulation) 2026

Ranked 1st of 55 teams in the simulation track with an event-for-event C++ rebuild of the ABIDES market simulator (see Publications). The win earned an additional team poster presentation at the Agenthon workshop at NeurIPS 2026.

Kaggle Bronze Medal · The Pokémon Company PTCG AI Battle Challenge 2026

Ranked 677th of 6,807 teams (top 10%) in Kaggle's simulation competition, developing and evaluating partially observable game-playing agents with replay-based behavioral cloning, belief-aware Monte Carlo planning, and self-play.

RESEARCH EXPERIENCE

Independent Researcher
03/2026 - Present

METU Applied NLP Group Ankara, TR

- Co-organize the SIGTURK 2027 Shared Task at the SIGTURK 2027 Workshop (ACL 2027) with Çağrı Toraman, Evren Ayberk Munis, and İlker Kesen. Built the Text/LLM track: 1,000 culturally grounded items in five Turkic languages (Turkish, Azerbaijani, Kazakh, Uzbek, Uyghur) across four tasks, selected from 1,750 candidates and grounded in 620 sources; wrote the evaluator, ran the baselines, and built the shared-task website.
- Contribute to GemmaTurk, a 1.6B-parameter Turkish LLM: built a Turkish YouTube caption corpus (1.34M documents, 6.1B tokens), generated synthetic instruction and reasoning data, ran mid-training data-mix experiments, and run continued pre-training with long-context extension to 64K tokens.
- Upcoming: TurkBench 2, an agentic extension of the TurkBench Turkish LLM benchmark, in collaboration with Turkcell.

AIOps Research Intern
08/2024 - 09/2024

Havelsan Istanbul, TR

- Built LSTM and random-forest prototypes for log-based anomaly detection on HDFS logs for predictive maintenance; the LSTM reached 0.96 F1 and 99% accuracy on the test set.

NLP Research Intern
07/2024 - 08/2024

ITU NLP Lab Istanbul, TR

- Ran experiments on financial LLMs across sentiment analysis, tool use, reasoning, forecasting, and RAG, feeding into the Kaira capstone.

PROFESSIONAL EXPERIENCE

Co-Founder & Technical Lead
08/2025 - 04/2026

Fiction Studios Ankara, TR

- Co-founded Fiction Studios at Bilkent Cyberpark; led technical development and coordinated three additional contributors. Built the backend of Meddah, a multilingual LLM-powered museum and tourism audio guide, and pitched it to customers and investors.

Junior AI Engineer
03/2025 - 07/2025

CyberQuote Istanbul, TR

- Built RAG pipelines that retrieve up-to-date context from fragmented internal sources and PDF financial reports, and agent pipelines for internal financial LLM tools. Also developed news-sentiment pipelines and web integrations at a Phillip Capital subsidiary.

SELECTED PROJECTS

Kaira
2025

Undergraduate Capstone: Financial Reasoning LLM

- Fine-tuned Qwen-based LLMs with GRPO, Unsloth, and LoRA, reaching 58.08% FinQA accuracy with Qwen3-4B, up from 47.48% with Qwen2.5-3B; ran FinQA and GSM8K experiments on reward design and financial reasoning. Developed an Ollama-based chat application.

Tobor
2024 - 2025

Autonomous Warehouse Robot Code · Video

- Implemented the box pickup/drop-off mechanism for a team-built ROS2 warehouse robot with navigation and YOLO perception; contributed to data annotation.

AIZheimer
2024

Machine Unlearning in Stable Diffusion Code

- Adapted Forget-Me-Not unlearning to Stable Diffusion 2.1, implementing attention controllers and testing loss objectives to erase target concepts while preserving unrelated generation.

Biting The Bytes
2024

Turkish Diacritic Restoration Code · Demo

- Led ByT5-small fine-tuning with PEFT/LoRA for byte-level Turkish diacritic restoration, reaching 96.18% word-level accuracy on the test set and tying for 2nd of 17 teams in the course's Kaggle competition; co-deployed the model on Hugging Face Spaces.

To-AI-or-Not-to-AI
2023

GPT Detector Code · Demo

- Built an ensemble of three fine-tuned RoBERTa detectors combined by majority vote, reaching 94.21% accuracy on the held-out test split of a 26,270-example dataset prepared with teammates.

ACADEMIC SERVICE

Shared Task Co-organizer, SIGTURK 2027 2026 - Present

SIGTURK 2027 Workshop @ ACL 2027

Lead, METU NLP Connect 2026 - Present

Expert interview series of the METU Applied NLP Group; co-hosted 4 episodes

TECHNICAL SKILLS

Languages Python / JavaScript/TypeScript / SQL / C/C++

Frameworks & Tools PyTorch / TensorFlow / Transformers / PEFT / LoRA / Unsloth / Scikit-learn / Pandas / NumPy / ROS2 / YOLO

Specialties NLP / LLMs / RAG / Machine Unlearning / Model Evaluation

COURSES / LANGUAGES

Deep Learning Specialization DeepLearning.AI · Coursera

AWS Cloud Technical Essentials AWS · Coursera

Turkish Native **English** TOEFL iBT 109/120 (November 8, 2025)

Interests: Coffee / Books / Photography / Anime & Manga